

AI For Healthcare

Week 1

Interesting concepts:

1. Computer vision and pixels

It is fascinating how an image is brought down to a simple grid with hidden numbers behind them and how those grids are connected to bits. Learning about the math behind it really gave me a new perspective of looking at images. It completely changed how I think about digital media now.

2. Panaroma stitching

I have been using panorama photos for a long time but never really thought about how they actually work. I used to believe it was just a continuous video, but it turns out it is a bunch of individual photos stitched seamlessly by a computer. It is fascinating how the software finds matching points between the frames to align and blend them together perfectly.

3. Brain and visual perception

This part confused me a lot; Even though our eyes actually capture a flat, 2-D image on the retina, the brain automatically adds depth, shadows, and the effects of color to it. It is amazing how our mind takes this flat data from two different angles and calculates the distance to let us perceive a 3-D world.

4. How images are recorded

Another interesting thing was, how cameras record images by using a filter. Instead of capturing the full image the colors are separated. And what we actually see is an addition of primary colors. It made me realize that digital colour is just a clever combination of math and lighting filters.

5. Interpolation and resizing

What caught my attention about resizing images is that the computer can't just stretch a photo; it actually has to guess the missing information. When you zoom in an image, it has to use math to fill in the new pixels based on the colours around them. Knowing that stretching a picture requires the computer to pretty much invent new pixels from scratch is pretty wild.

Types of Healthcare Reports

1. **Laboratory Reports:** These show the raw results of fluid tests compared against normal ranges. They help give a comparison and check if the body is functioning properly.
2. **Pathology Reports:** These are written after a specialist examines a tissue sample, like a biopsy, under a microscope. They give a highly detailed analysis to find out if a disease or tumor is cancerous or benign.
3. **Operative Reports:** This is a step-by-step log written by a surgeon immediately after a surgery is finished. It details exactly what procedures were done, what tools were used, and how the patient responded.
4. **Discharge Summaries:** This report gives a complete overview of everything that happened during a patient's hospital stay. It includes the reason they were admitted, the treatments they received, and instructions for their care at home.

Medical Imaging

Medical imaging is the use of technology to create visual representations of the inside of the human body for diagnosis, monitoring, and treatment of diseases or injuries. It helps healthcare professionals detect abnormalities without invasive procedures.

Types of medical imaging:

1. X-Ray
2. Computed Tomography (CT) Scan
3. Magnetic Resonance Imaging (MRI)
4. Ultrasound
5. mammography
6. Positron Emission Tomography (PET) Scan